

Instructor key

Fictional submissions - reveal only after independent grading

These are designed scores under the workshop rubric, not real grades or a guarantee that an independent grader will choose identical partial credit. All answers were newly composed from the assignment and checked reference calculations; no real student answers were used. Character names are fictional labels, not evidence for grading.

Keep this key outside the grading input folder. Grade the four PDFs first, then reveal this key. To demonstrate anonymous reporting, ask for Submission A-D in the report and keep the Marvel names only in a separate identity mapping.

Problem totals: P1 20, P2 20, P3 15, P4 10, P5 15, P6 20 (100 overall). The criterion-level rubric is in rubric.json. Accept equivalent correctly labeled spectral units and reasonable rounding. P6 assumes a nadir-facing face, a uniformly emitting Earth disk and grey thermal absorptivity equal to emissivity. Other justified geometry conventions require instructor judgment.

For the P6 emissivity criterion (6 points): award 1 point for each correct absorbed-plus-internal power and 2 points for each correct emissivity, with valid supporting work. For P5, component derivations and the final total are separate criteria; retain correct component credit when the final result is wrong. All other criteria here are fully met or clearly omitted/incorrect.

Tony Stark: 100/100

P1: 20/20 - submission page 1

frequency Jacobian and result: 5/5; wavenumber Jacobian and result: 5/5; wavelength inverse derivation: 5/5; wavenumber inverse derivation: 5/5

All four transformations/inverses are derived with consistent spectral conventions.

P2: 20/20 - submission page 1

10 μm brightness temperature with unit conversion: 8/8; 0.7 μm radiance: 4/4; 1000 cm^{-1} radiance: 4/4; 31.4 GHz radiance: 4/4

All results and spectral-density conversions are correct.

P3: 15/15 - submission page 2

both radar wavelengths and wavenumbers: 6/6; 650 nm photon energy in eV: 4/4; 8 to 12 μm window endpoints and units: 5/5

All conversions and units are correct.

P4: 10/10 - submission page 2

solar dilution and projected/emitting areas: 4/4; effective temperature: 3/3; greenhouse explanation: 3/3

Correct energy balance, temperature, and greenhouse explanation.

P5: 15/15 - submission page 3

downward and upward flux: 6/6; direct actinic contribution: 3/3; reflected actinic contribution: 3/3; total actinic flux: 3/3

Correct projected and actinic fluxes.

P6: 20/20 - submission page 3

solar/internal heating and six-face emission balance: 4/4; two emissivities: 6/6; Earth irradiance geometry: 3/3; thermal absorptivity equals emissivity: 3/3; revised energy balance: 2/2; revised temperatures: 2/2

Complete solution with explicit geometry and thermal absorptivity assumptions.

Peter Parker: 75/100

P1: 15/20 - submission page 1

frequency Jacobian and result: 5/5; wavenumber Jacobian and result: 5/5; wavelength inverse derivation: 5/5; wavenumber inverse derivation: 0/5

Wavenumber inverse omitted: -5.

P2: 16/20 - submission page 1

10 um brightness temperature with unit conversion: 8/8; 0.7 um radiance: 0/4; 1000 cm⁻¹ radiance: 4/4; 31.4 GHz radiance: 4/4

Optical result is mislabeled per micrometre, a factor of 1e6 error: -4.

P3: 15/15 - submission page 2

both radar wavelengths and wavenumbers: 6/6; 650 nm photon energy in eV: 4/4; 8 to 12 um window endpoints and units: 5/5

All conversions correct.

P4: 7/10 - submission page 2

solar dilution and projected/emitting areas: 4/4; effective temperature: 3/3; greenhouse explanation: 0/3

Surface-to-orbit distance does not explain the greenhouse temperature difference: -3.

P5: 12/15 - submission page 3

downward and upward flux: 6/6; direct actinic contribution: 3/3; reflected actinic contribution: 3/3; total actinic flux: 0/3

Components are correct, but their final sum is wrong: -3. Do not also penalize the reflected component.

P6: 10/20 - submission page 3

solar/internal heating and six-face emission balance: 4/4; two emissivities: 6/6; Earth irradiance geometry: 0/3; thermal absorptivity equals emissivity: 0/3; revised energy balance: 0/2; revised temperatures: 0/2

Both emissivities correct; entire Earth-heating extension omitted: -10.

Natasha Romanoff: 50/100

P1: 10/20 - submission page 1

frequency Jacobian and result: 5/5; wavenumber Jacobian and result: 5/5; wavelength inverse derivation: 0/5; wavenumber inverse derivation: 0/5

Spectral conversions correct; both inverse functions replaced by reciprocals: -10.

P2: 8/20 - submission page 1

10 um brightness temperature with unit conversion: 8/8; 0.7 um radiance: 0/4; 1000 cm⁻¹ radiance: 0/4; 31.4 GHz radiance: 0/4

Correct brightness temperature; all three emitted-radiance calculations omitted: -12.

P3: 10/15 - submission page 2

both radar wavelengths and wavenumbers: 6/6; 650 nm photon energy in eV: 4/4; 8 to 12 um window endpoints and units: 0/5

Radar and photon conversions correct; window reciprocals use inconsistent units: -5.

P4: 7/10 - submission page 2

solar dilution and projected/emitting areas: 4/4; effective temperature: 3/3; greenhouse explanation: 0/3

Balance and temperature correct; clouds do not create energy and greenhouse mechanism is missing: -3.

P5: 9/15 - submission page 3

downward and upward flux: 6/6; direct actinic contribution: 3/3; reflected actinic contribution: 0/3; total actinic flux: 0/3

Reflected actinic flux lacks the factor 2: -3; final total consequently wrong: -3 under the separate final-result criterion. Preserve all direct/irradiance credit.

P6: 6/20 - submission page 3

solar/internal heating and six-face emission balance: 4/4; two emissivities: 2/6; Earth irradiance geometry: 0/3; thermal absorptivity equals emissivity: 0/3; revised energy balance: 0/2; revised temperatures: 0/2

Correct six-face balance: 4/4. Emissivities: 2/6 for both correct absorbed powers, but neither numerical emissivity is correct; six-face factor lost in substitution. Earth extension: 0/10.

Thor Odinson: 30/100

P1: 5/20 - submission page 1

frequency Jacobian and result: 5/5; wavenumber Jacobian and result: 0/5; wavelength inverse derivation: 0/5; wavenumber inverse derivation: 0/5

Frequency conversion correct; wavenumber Jacobian missing and both inverses incorrect: -15.

P2: 8/20 - submission page 1

10 um brightness temperature with unit conversion: 8/8; 0.7 um radiance: 0/4; 1000 cm⁻¹ radiance: 0/4; 31.4 GHz radiance: 0/4

Brightness temperature correct; three radiances omitted: -12.

P3: 6/15 - submission page 2

both radar wavelengths and wavenumbers: 6/6; 650 nm photon energy in eV: 0/4; 8 to 12 um window endpoints and units: 0/5

Radar conversions correct; photon calculation confuses wavelength/frequency and energy units, and window conversion is wrong: -9.

P4: 4/10 - submission page 2

solar dilution and projected/emitting areas: 4/4; effective temperature: 0/3; greenhouse explanation: 0/3

Correct global energy balance earns 4; temperature unevaluated and explanation incorrect: -6.

P5: 3/15 - submission page 3

downward and upward flux: 0/6; direct actinic contribution: 3/3; reflected actinic contribution: 0/3; total actinic flux: 0/3

Correct direct actinic flux earns 3; projected fluxes wrong and reflected/total actinic answers omitted: -12.

P6: 4/20 - submission page 3

solar/internal heating and six-face emission balance: 4/4; two emissivities: 0/6; Earth irradiance geometry: 0/3; thermal absorptivity equals emissivity: 0/3; revised energy balance: 0/2; revised temperatures: 0/2

Correct initial six-face balance earns 4; remaining work omitted: -16.